# Experiment-10: A* Search Algorithm using Python

**Aim**

To develop a Python program to implement the **A* (A-Star) Search Algorithm** for finding the shortest path between a start node and a goal node using path cost and heuristic values.

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**Algorithm**

1. Create the graph with nodes and edge costs.
2. Define the heuristic value **h(n)** for each node.
3. Initialize:
    - Open list with the start node.
    - Closed list as empty.
4. Select the node with the lowest **f(n) = g(n) + h(n)**.
5. If the selected node is the goal, stop and display the path.
6. Otherwise, expand its neighboring nodes.
7. Update the cost and parent of each neighbor if a better path is found.
8. Repeat Steps 4–7 until the goal is reached or the open list becomes empty.
9. Display the shortest path and total cost.

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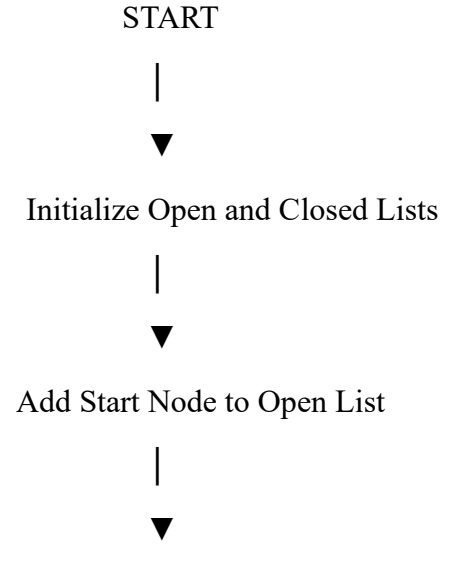
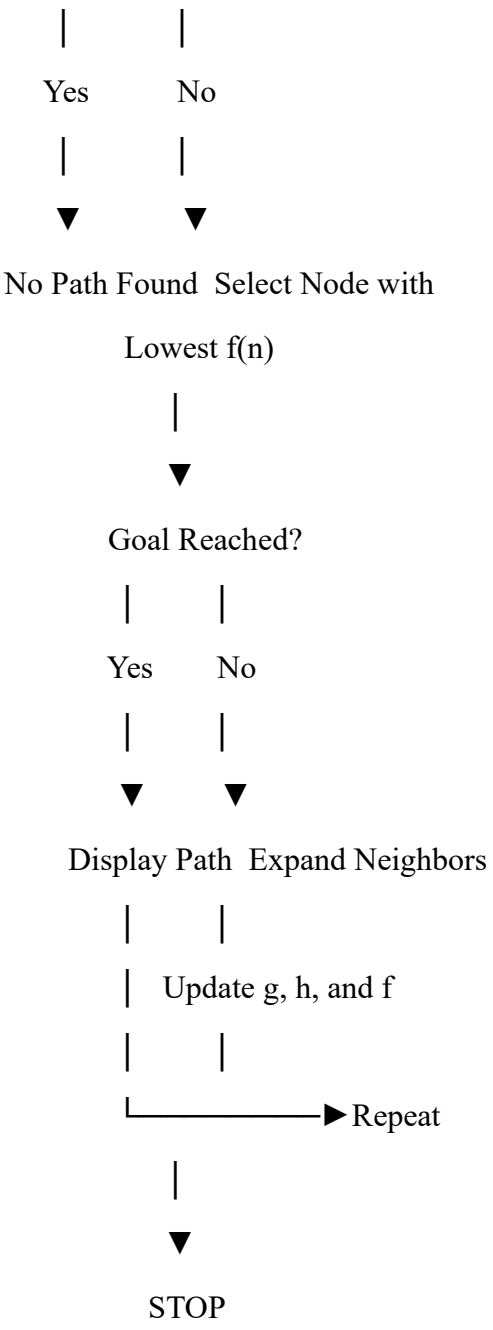
**Flowchart**



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**Python Code**

```
# A* Search Algorithm


graph = {
    'A': [('B', 1), ('C', 3)],
    'B': [('D', 3), ('E', 6)],
    'C': [('F', 5)],
```

```
    'D': [('G', 4)],
    'E': [('G', 2)],
    'F': [('G', 2)],
    'G': []
}

# Heuristic values
heuristic = {
    'A': 7,
    'B': 6,
    'C': 4,
    'D': 4,
    'E': 2,
    'F': 1,
    'G': 0
}

def astar(start, goal):

    open_list = {start}
    closed_list = set()

    g = {start: 0}
    parent = {start: start}

    while open_list:

        node = None
```

```
for v in open_list:
    if node is None or g[v] + heuristic[v] < g[node] + heuristic[node]:
        node = v

if node == goal:
    path = []

    while parent[node] != node:
        path.append(node)
        node = parent[node]

    path.append(start)
    path.reverse()

    print("Shortest Path:", path)
    print("Total Cost:", g[goal])
    return

for (neighbor, weight) in graph[node]:

    if neighbor not in open_list and neighbor not in closed_list:
        open_list.add(neighbor)
        parent[neighbor] = node
        g[neighbor] = g[node] + weight

    elif g[neighbor] > g[node] + weight:
        g[neighbor] = g[node] + weight
        parent[neighbor] = node
```

```
            if neighbor in closed_list:
                closed_list.remove(neighbor)
                open_list.add(neighbor)

        open_list.remove(node)
        closed_list.add(node)

    print("Path does not exist.")

astar('A', 'G')
```

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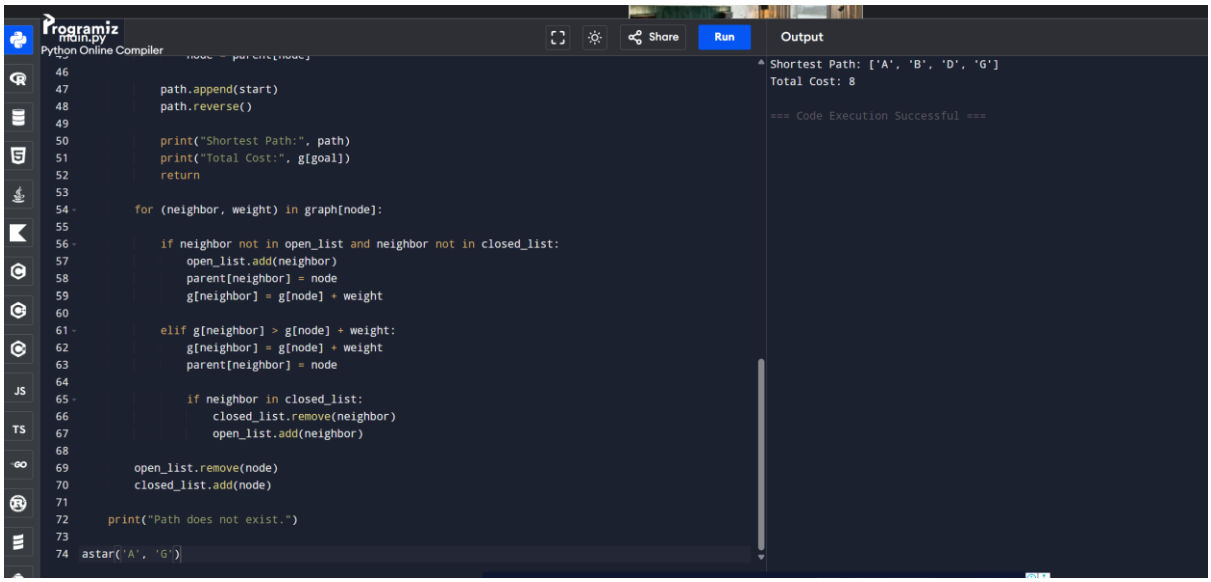
**Sample Output**

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**Result**

The **A* Search Algorithm** was successfully implemented in Python. The algorithm used the evaluation function **f(n) = g(n) + h(n)** to efficiently find the shortest path from the start node to the goal node.

- **Shortest Path: A → B → D → G**
- **Total Cost: 8**